Travelling Salesperson Problem

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1 Introduction

1.1 Problem definition

Travelling Salesperson Problem (TSP) is an optimisation problem that can be presented as follows. There is a given graph $G = (V, E)$ such that $\forall_{v,w \in V} \ \exists_{e \in E}$ (complete graph). Every $e \in E$ has an assigned non-negative weight. To solve the TSP problem we want to find the smallest Hamiltonian cycle, that is:

1. **Hamiltonian cycle** – a cycle that visits every $v \in V$ exactly once and returns to the starting vertex.
2. For a cycle $(v_1, v_2, \ldots, v_n)$ the **smallest Hamiltonian cycle** is the one that minimises $\sum_{i=1}^{n-1} w(v_i, v_{i+1}) + w(v_n, v_1)$.

1.2 Number of Hamiltonian cycles

Since G is a complete graph there are $\frac{(n-1)!}{2}$ distinct Hamiltonian cycles (fixing one starting vertex removes the n-fold rotational symmetry; dividing by 2 removes the direction symmetry for undirected graphs).

2 Datasets

All experiments were run on five benchmark instances from the TSPLIB World TSP collection (`https://www.math.uwaterloo.ca/tsp/world/countries.html`). Edge weights are Euclidean distances rounded to the nearest integer (EUC_2D convention).

Instance	Cities	Region
wi29	29	Western Sahara
dj38	38	Djibouti
qa194	194	Qatar
uy734	734	Uruguay
zi929	929	Zimbabwe

3 Task 1 – Random permutation sampling

1000 random Hamiltonian cycles were generated for each instance. Table 1 reports:

(a) the average of the per-group minimum over 100 groups of 10 draws,

(b) the average of the per-group minimum over 20 groups of 50 draws,

(c) the global minimum across all 1000 draws.

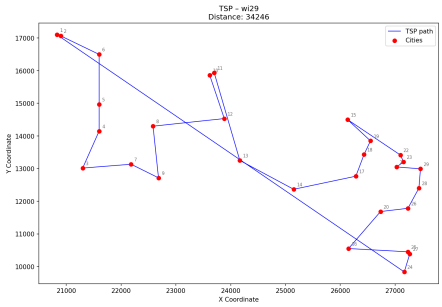
As expected, larger groups yield a lower average minimum because the chance of sampling a good tour at least once per group increases.

Table 1: Random sampling results (1000 permutations per instance).

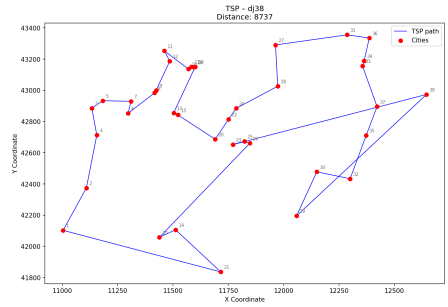
Instance	Avg min (groups of 10)	Avg min (groups of 50)	Global min
wi29	94 709	87 309	76 356
dj38	24 784	23 367	21 642
qa194	87 889	85 506	81 121
uy734	1 597 079	1 579 043	1 565 501
zi929	2 307 289	2 279 280	2 230 126

4 Task 2 – Visualisation of the best random tour

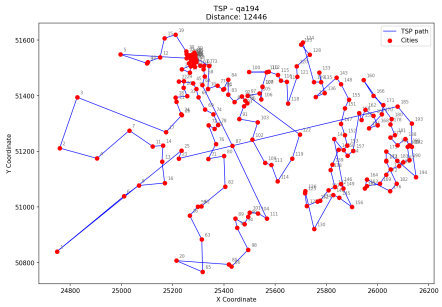
Figures 1a–1e show the best tour found by random sampling for each instance. Even visually it is clear that random tours contain many *crossing* edges – a hallmark of suboptimal Euclidean solutions. In any optimal Euclidean tour no two edges can cross, because uncrossing them (2-opt move) strictly reduces the total length by the triangle inequality.



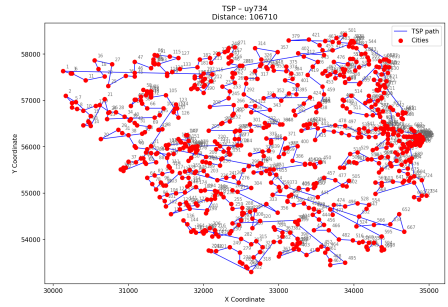
(a) wi29 – best random tour (76 356)



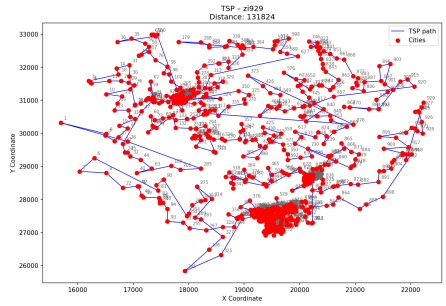
(b) dj38 – best random tour (21 642)



(c) qa194 – best random tour (81 121)



(d) uy734 – best random tour (1 565 501)



(e) zi929 – best random tour (2 230 126)

Figure 1: Best tours found by random sampling.

5 Task 3 – Minimum Spanning Tree

5.1 MST weights

Table 2 lists the MST weight computed by Prim's algorithm alongside the best random tour for comparison.

Table 2: MST weights vs. best random tour.

Instance	**MST weight**	**Best random tour**
wi29	21 753	76 356
dj38	5 828	21 642
qa194	8 026	81 121
uy734	69 873	1 565 501
zi929	82 863	2 230 126

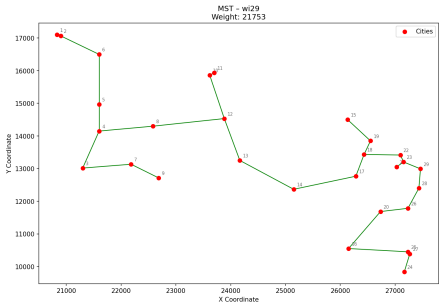
5.2 Why $w(\text{MST}) \leq w(\text{OPT})$

Let T^* be an optimal TSP tour. Removing any single edge from T^* yields a spanning tree of G. Since the MST has the minimum possible weight among all spanning trees, we have:

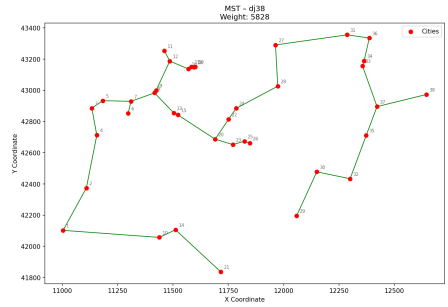
$$w(\text{MST}) \;\leq\; w(T^* \setminus \{e\}) \;<\; w(T^*)$$

Therefore the MST weight is a lower bound on the optimal tour length.

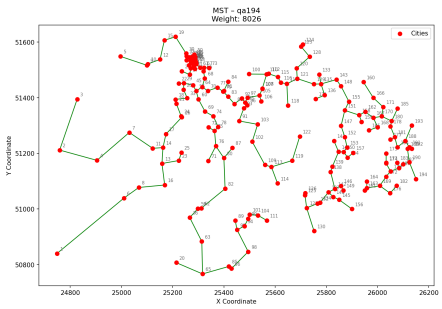
Figures 2a–2e show the MST for each instance.



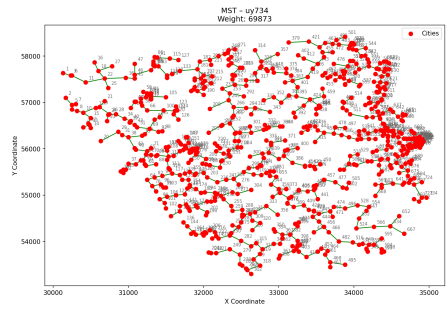
(a) wi29 – MST (weight 21 753)



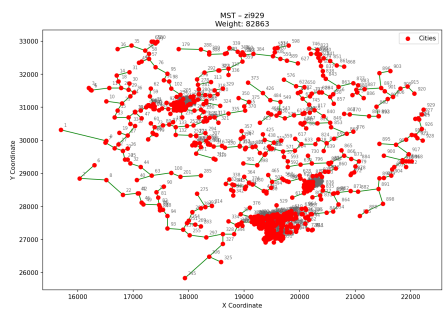
(b) dj38 – MST (weight 5 828)



(c) qa194 – MST (weight 8 026)



(d) uy734 – MST (weight 69 873)



(e) zi929 – MST (weight 82 863)

Figure 2: Minimum Spanning Trees computed by Prim's algorithm.

6 Task 4 – MST-based 2-approximation

6.1 Algorithm

A Hamiltonian cycle is built from the MST by performing a depth-first search (DFS) pre-order traversal and listing each city the first time it is visited. The resulting sequence forms a valid tour.

6.2 Results

Table 3: MST-based tour vs. MST weight (2-approximation guarantee).

Instance	**MST weight**	**MST-TSP tour**	**Ratio**	$\leq 2\times$ **MST?**
wi29	21 753	34 246	1.574	
dj38	5 828	8 737	1.499	
qa194	8 026	12 446	1.551	
uy734	69 873	106 710	1.527	
zi929	82 863	131 824	1.591	

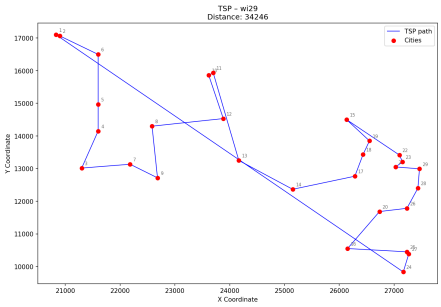
6.3 Why the tour is at most $2 \times w(\text{MST})$

A full DFS walk of the MST traverses every edge exactly twice, giving a closed walk of total length $2 \cdot w(\text{MST})$. When repeated cities are shortcut (skipped) the triangle inequality guarantees the shortcut path is no longer. Therefore the resulting Hamiltonian tour satisfies:

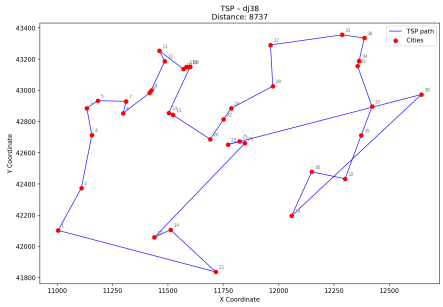
$$w(\text{MST-TSP}) \;\leq\; 2 \cdot w(\text{MST}) \;\leq\; 2 \cdot w(\text{OPT})$$

which is the classical *2-approximation* for metric TSP. All five instances confirm the bound, with ratios between 1.50 and 1.59 – well within the theoretical guarantee.

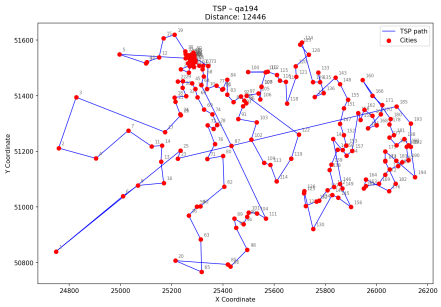
Figures 3a–3e show the resulting tours.



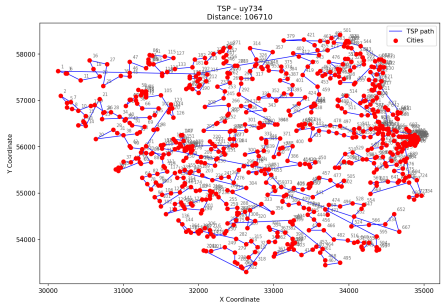
(a) wi29 – MST-based tour (34 246)



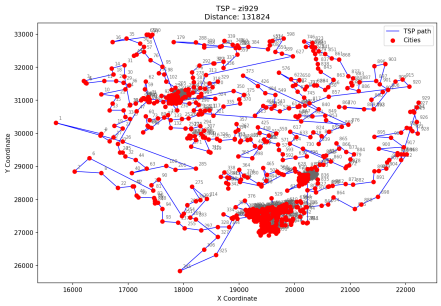
(b) dj38 – MST-based tour (8 737)



(c) qa194 – MST-based tour (12 446)



(d) uy734 – MST-based tour (106 710)



(e) zi929 – MST-based tour (131 824)

Figure 3: TSP tours derived from the MST via DFS pre-order.

7 Conclusions

Random sampling produces tours that are many times longer than the MST-based approximation – roughly 20× longer on the larger instances – which confirms that pure random search scales extremely poorly. The MST-based 2-approximation, by contrast, consistently produces tours within a factor of 1.6 of the MST lower bound across all five datasets, demonstrating the practical value of structure-guided construction heuristics.